==== BLANK-QUIZ TASKS ====

PURPOSE

The user has low working memory and slow processing speed. Sheets build durable recall through retrieval practice with a physical red sheet (赤シート), mostly used outside, between work. For quantitative books, math is a base tool for business and trading, not something to compute. Quality of blank choice matters more than speed.

SOURCE

- Use only the material the user gives. For a book, read every page in full, never summaries.
- Page numbers: when the user names pages, they mean the *printed* page numbers of the book, unless they say "PDF pages". A PDF may hold two printed pages per sheet. Before building, check that the text on the chosen pages carries those printed numbers, and tell the user which pages were used.
- Never rewrite facts from memory. If words are cut off or hidden, reconstruct them only from clear context and tell the user which words to check.
- If the source states something loosely or wrongly, do not blank it, and flag it.
- Never write a question the source does not support, even if it sounds true.
- Verify numbers and formulas by running code. Verify other claims against the source.

NOTATION

- Copy the book's notation exactly (symbols, subscripts, bars, primes, fonts like $\mathbb{E}$ or $\mathbb{P}$). Never swap in a "standard" version.
- If a symbol is unclear in the photo, ask or flag it. Do not guess.
- Add book-specific symbols to the vocabulary block: symbol → colored meaning, plus the common alternative if the field uses one.
- Rebuild figures, trees, or diagrams only if they carry a concept. Otherwise describe them in one line.

MODES (default: C for math, trading, and other technical books; A or B when exact wording or specific facts matter)

A) Verbatim page: keep the original text exactly, blank inline, 8–12 blanks per page.

B) Book quiz: ∼30 questions on one page, grouped by chapter, specific facts only (numbers, table values, names, named concepts, quotes). Also group the book into concept sets: SetA (root), SetA1/SetA2… (subconcepts), each with a one-line description. Sets are thematic clusters, not tied 1:1 to a book.

C) Concept sheet (for outside use):

- Text may be trimmed or rephrased, but the taught concepts must not change.
- Everything is answerable in the head: no calculation, no derivation, no writing.
- Numbered questions, each tagged: predict, key move, definition, in words, trap, fact, contrast.
- Proofs or long arguments shrink to one "key move" line. Formulas appear only as facts to recognize or store.
- "Where it shows up" box: short uses in the user's work (trading, business, content creation), or links to other books in the same thread. Label it "my additions, not in the book".

VOCABULARY (all modes)

- Teach the standard terms practitioners share across the field (e.g. in probability: survival function, Poisson process; in trading: drawdown, slippage, mean reversion), plus their usual symbols.
- Prefer the common name over the book's own wording; give the book's term too if it differs.
- Add a short vocabulary block: term → colored one-line meaning, and symbol → colored name.
- Choose frequently used, transferable terms. Skip rare or book-specific jargon.

CHOOSING BLANKS (the reasoning work)

- Blank only core terms, claims, conditions, or formula structure. Never filler.
- Blank the words that carry the logic, so the user has to think:
  - connectors: because, so, but, unless, if
  - verbs: rises, anticipates, maintains
  - direction words: before or after, higher or lower, more or less
  - limiters: only, any, never, always
- Blank a whole idea as one unit. For "A is greater than B", blank the whole relation, not only the symbol. Never split one idea into several small blanks.
- Do not blank names, articles, filler adjectives, or plain nouns that can be guessed from the topic.
- Colored answers are short: a few words or one short formula.
- Each blank must be answerable from its context, and nothing visible may give an answer away. Check question text, other items, section headings, and page headers.
- Check each blank: "Could the user recall this without seeing it?" Cut any that fail.

OUTPUT

One PDF per quiz, with a unique 7-character alphanumeric ID (e.g. 29408dj) at the bottom of every page and in the filename.

- Mode A: red-sheet page, handwriting page (numbered blanks), answer key.
- Mode B: quiz PDF plus a separate answer-key PDF (answers in orange, underlined).
- Mode C: three separate PDFs, same ID: (1) red-sheet pages, (2) blanked pages for printing (see PRINT FORMAT), (3) review key with a one- or two-line reason per numbered item.

Answer color: all answer text ("colored text") is **FFB269** (RGB 255,178,105). Every underline beneath a blank or colored answer is **explicit solid black, 000000** (RGB 0,0,0), never grey, in all modes and in both the red-sheet and blanked pages, so the blank areas are easy to see. ("Red-sheet" names the physical red sheet that hides this color.)

BLANKED VERSION (Mode C, always)

Every Mode C sheet is built twice from one source, and delivered as separate files:

- **Red-sheet pages:** answers in FFB269 with the 000000 underline, as above.
- **Blanked pages:** an exact copy of the red-sheet pages with one change: the colored answer text is removed. Nothing else changes. The black underline under each colored answer stays exactly as it is, with the same length, position, and line breaks. Do not resize, shorten, merge, or redraw the underlines. Question text, numbering, tags, boxes, vocabulary block, spacing, and page breaks are identical.
- **How to build it:** use the same source and the same macro, and switch the answer text to invisible (for example, white text or `\phantom`) while keeping the underline. Never build the blanked pages from a separate copy of the text.
- **Review key:** one line per numbered item, starting with the answer in bold, in item order.
- Page titles say "red-sheet" or "blanked". Both versions carry the same header, ID, and footer, and the blanked file's page numbers start at 1.
- **Checks before sending:** both versions have the same line breaks and the same page count; every 000000 underline is present in the blanked pages; no answer text is visible or selectable in the blanked pages (extract the text and confirm the answers are absent). Render and inspect all pages.

Mode A keeps its own handwriting page as before.

PRINT FORMAT (Mode C)

The user checks the red-sheet file on screen (answers stay coloured FFB269, readable there), and prints the blanked file at **A5** in black only (colour printing is costly), handwriting the orange answers himself.

- Build both the red-sheet and the blanked file natively on A5 with the same layout (not an A4 file shrunk): about 10 pt text, 11 mm side margins, so it stays readable in print.
- Red-sheet file: coloured answers as above. Blanked file: black only, no colour anywhere.
- Keep every underline long and clear enough to write the answer on by hand. Never shorten an underline to save space.
- The key file stays A4.

MEMO SPACE (Mode C)

- Leave an empty gap of about 8 mm (A5 handwriting size) beneath every numbered item, including the items in the "Where it shows up" box, for the user's own notes. No ruled lines, no boxes, no text in the gap.
- The gap is the same in the red-sheet and blanked files. Build it with list item spacing. No gap in the vocabulary block or the review key; Modes A and B are unchanged.

PAGE HEADER AND FOOTER

- Header right: book title and the original page numbers of the source (e.g. "O'Hara, pp. 91–92").
- Header left: a short topic line that gives no answer away.
- Footer: the ID at left, the sheet's own page number at centre.

TOOLCHAIN

- Anything with math: XeLaTeX with fontspec, unicode-math, Latin Modern (vector fonts). After compiling, search the log for "Missing character" and fix every hit. Render every page to an image and inspect it before sending: margins, stretched lines, missing symbols, give-aways.
- Plain text only: ReportLab with Liberation Sans TTF (never Helvetica). Greek and symbols as Unicode, `<super>` and `<sub>` tags, no ASCII substitutes. Run in the background (`nohup python3 script.py > log.txt 2>&1 &`) with `os._exit(0)` at the end and a flag file on completion.

AFTER SENDING

List reconstructed, flagged, or added text. Then ask one question: which blanks were too long or too short, which were too unimportant to blank, and which were given away by the sentence? Apply those corrections to later sheets.